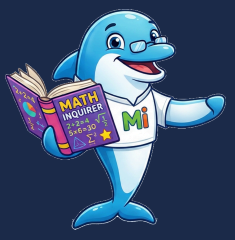
TUTOR GUIDE • LEVEL 1 • TERM 4 • LESSON 3 OF 10

Copy on the Grid



Big idea: A grid lets us say exactly where things are and copy them exactly. On a grid of dots we count the steps between the dots. Then we can say exactly where to go and copy a figure exactly.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Geometry: 2D shapes, copy figures on a dot or square grid (Primary 1, p. 32) (Copy figures on a dot or square grid (Primary 1, Geometry, 1.4, p. 32); Position words: left, right, up and down, and counting steps between dots (Primary 1 process skills); Listing every case without missing one (Primary 1 heuristics, pp. 13 to 14)) Strand label for the tutor only. Copying figures on a dot or square grid is a listed Primary 1 outcome (Geometry, 1.4). Coordinates are not in the syllabus anywhere, so the lesson says only '3 right, 2 up' from a start dot, never a pair of numbers. Start here is Primary 1 work with scaffolding: copy a figure by counting the steps between the dots, say a move as right and up, follow directions with right and up, pick the exact copy, finish a rectangle and read a path. BEYOND PRIMARY 1 ON PURPOSE at Think harder and Go further only: counting how many different routes there are on a grid when a counter moves only right or up (how many ways, a listing and counting move), a path that repeats (continue a pattern), the left move, adding at the corners to count the routes on a bigger grid, and a route that must pass a given corner. Named puzzle: Count the Routes. Owner decisions, 22 and 23 Oct 2026: strong content with scaffolding, said plainly to families; every lesson names a beyond item; Start here stays approachable. Do not describe the lesson as MOE-aligned. Wording rule (Term 1 audit, pre-flight D18): wrong working is always 'a child' or 'a child in another class', never a cast name. There is no school-paper evidence for grid copying (P1 paper patterns, section 6), so the wording is unverified. Non-routine types (tutor only): how many ways (routes in a small grid), continue a path from its directions.
Objectives	• Copy a figure exactly on a dot grid by counting the steps between the dots, not the dots. • Say a move from a start dot as how many right and how many up, and follow directions with right and up. • Pick the exact copy of a figure and finish a rectangle from three corners. • Think harder: list every route that goes only right or up on a small grid, continue a repeating path, and follow a path with a left move. • Go further: add the corner numbers to count the routes on a bigger grid, count the routes that pass a given corner, and describe a figure for a friend to copy.
Joining cold	**Remember warm-up covers:** Rows and columns of counters (Term 2 Lesson 5, taught again with a picture); Steps between dots in a row (Term 1 Lesson 6, jumps, taught again with a picture); Right and up on a grid of squares (Term 1 Lesson 11, position words, taught again with a picture). Only counting to 10 and knowing which way is right and which way is up. Nothing from the lessons before is needed: the warm-up re-teaches rows and columns, steps between dots, and right and up on a grid. The words step, route and exact copy are taught in Explore and the Mini-lesson. Fairly safe to miss: the next lesson does not need it, but Mi's Challenge Quest in the last lesson of the year has a gate on grids and routes. A child who misses it watches the recording and does workbook pages 1 and 2. A child who joins here cold can follow: the warm-up re-teaches rows, steps and right and up, and Start here keeps to grids up to 5 steps across.

Child materials	• Your Term 4 book (Book 7), workbook pages 1 to 5, and the grid page (page 6). The grid page is not cut: it has 3 route grids and 2 copy grids to draw on • A counter to walk along the grid: a button or a small coin. A button is best • A ruler for straight lines (the edge of a card works too) • No printer? Draw rows of dots on paper: 4 dots across and 4 dots down for each route grid • A pencil, a rubber and spare paper
Tutor materials	• The slide deck • Your own grid page and a button to show on camera, if you wish • A visible clock Nothing needs to be bought. A button or small coin is a small piece: it stays on the table; nothing goes in the mouth or ears; keep it away from younger brothers and sisters. No matches, toothpicks or anything sharp. Nobody cuts during the lesson: the grid page is not cut. Children draw with a pencil; they draw straight lines on dots, with a ruler or the edge of a card if they like.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Beyond Primary 1 (tutor only)	Count the Routes at T1, T2, G1, G2: Counting how many different routes a counter can take from the bottom left to the top right of a grid when it moves only right or up (a grid 3 across and 1 up has 4, 2 across and 2 up has 6, 3 across and 2 up has 10), by listing every route in order and then by adding at the corners, and counting the routes that pass a given corner. Counting ways is not a Primary 1 outcome (the nearest idea is listing every case, a heuristic) and routes are not in the syllabus at any level. The other items (a repeating path and a left move) are following-directions tasks. Named puzzle: Count the Routes. Never in Start here, never advertised to children.
Contest-style move	Used at Think harder, Go further and the Investigation: count the routes: right or up only, listing every route in order so none is missed (how many ways); continue a path from its directions (a repeating pattern); a route that must pass a given corner (join two counts); add at the corners to count the routes on a bigger grid. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first. Check the answers: S1 a copy, 3 and 3; S2 2 right, 3 up; S3 a line 4 right then 1 up; S4 B; S5 3 right, 2 up; S6 True, False, False, True; T1 U R R R, R U R R, R R U R and 4; T2 R R U U, R U R U, R U U R and 6; T3 9 and 3, 15 and 5; T4 2 right, 2 up; G1 4; 3, 6, 10 and 10; G2 3, 2 and 6; G3 any three corners; I1 5 and 5, every route is 3 right and 2 up.
- Tell parents a week before: the grid page (page 6 of the book) is not cut and needs a button from home. For the lesson after this one, the 4 paper squares on the fold-squares page of Book 7 are torn out and cut by a grown-up, and the family finds 12 small blocks of one size (building bricks or dice); a family with no blocks is fine, the lesson works with pictures.
- Have the slides open. Have your own grid page and a button to show on camera if you wish. Practise the body directions for the break so that right and left match what children see on the screen.
- Check the recording captures the tutor and the shared screen. Children's video is excluded; any unmuted voice is recorded. Greet the whole group first, say the notice, then start the recording; avoid first names while it runs.
- Be plain with families: copying a figure on a dot grid, saying a move as right and up and following directions are Primary 1 work; counting how many different routes there are, adding at the corners and a route that must pass a corner go beyond Primary 1 on purpose. Coordinates are not in the syllabus: say '3 right, 2 up', never a pair of numbers. Do not call the lesson MOE-aligned.
- Wording rule: say step and jump for the gap between two dots; never count the dots. Wrong working belongs to 'a child' or 'a child in another class', never to a cast name and never to a child in the group. Slides keep the answers off the screen while you ask: read every question with its numbers aloud.

Safeguarding and recording (this lesson)

- A button or small coin is a small piece. It stays on the table: nothing goes in the mouth or ears. No matches or toothpicks. Nobody cuts during the lesson; the grid page is not cut.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a drawing up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Wrong working belongs to 'a child' or 'a child in another class', never to a child in the group, and never to a name. The names in the stories (Ali, Mei, Ravi, Ben) are for the story only. Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Moves are gentle; a child may stay seated and move one arm.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Copy on the Grid	1:36
2–8	**Remember warm-up**	6	3: Warm-up: rows and columns, Steps between dots, Right and up	4:16
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:12
12–23	**Explore**	11	5: Explore: copy it exactly, A copy and the figure, Explore: walk your button, The line your button walked, What did we notice?	8:36
23–34	**Mini-lesson**	11	10: Steps, not dots, 4 dots, 3 steps, Copy corner by corner, A, B and C, Say where to go, 4 right, 2 up, Follow the directions, Where did we stop?, Different routes, Three routes, no more	10:17
34–38	**Brain break**	4	1: Counter break	3:16
38–70	**Workbook**	32	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	24:36
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:26
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:47
82–85	**Look-back**	3	1: Look-back	2:06
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:36
	Total	**90**		**67:43**

The script needs 67:43 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 80:03. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 38 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 38, skip in this order: Start with: shorten Explore round 3 (what did we notice) to one question; Then: cut the your-turn in the corner-by-corner example (s-m2); Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:36 of 2:00

0:00	SLIDE	Show slide **Copy on the Grid** (s-title)
0:10	DO	Greet the whole group warmly, before the recording starts. Check that you can see each child, or ask the grown-up. [0:12]
0:22	SAY	Hello, inquirers! Pencil ready, and your grid page and your button on the table. Cameras can be on or off, as you like.
0:34	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is not muted. You may keep your microphone and camera off.
0:51	DO	Start the recording now. Do not use children's first names on the recording. [0:05]
0:56	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your grid page and a button. Pause or rewind any time.*
1:08	SAY	Microphone off until I ask you to speak. To answer, say it with me on three, show fingers, or give a thumbs up. Nobody holds paper up to the camera.
1:23	SAY	Today a grid helps us say exactly where things are. We copy a figure exactly, we say where to go, and we count different routes.
	NOTE	Say the recording statement and the microphone routine in the same words every lesson, for anyone who joined late. Say the notice before you start the recording, and do not use children's first names once it is running. The routine and the four inquirer rules were introduced earlier in the year: do not re-teach them.
	NOTE	Plain statement for families when it comes up: copying a figure on a dot grid and saying a move as right and up are Primary 1 work; counting how many different routes there are, adding at the corners and a route that must pass a corner go beyond Primary 1 on purpose. There is no school-paper evidence for grid copying, so the wording is unverified.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. A child without the grid page draws rows of dots on paper: 4 dots across and 4 dots down.

2:00 – 8:00 Remember warm-up

script 4:16 of 6:00

2:00	SLIDE	Show slide **Warm-up: rows and columns** (s-rem1)
2:12	REC	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*
2:20	SAY	Counters in a grid. Rows go across. Columns go down.
2:26	ASK	How many rows are there? How many counters are in each row? [wait about 25 s for answers]
2:56	CHILDREN	Your turn. Two rows with four dots in each row. How many dots in all? Say it with me. [0:22]
3:18	SLIDE	Show slide **Steps between dots** (s-rem2)
3:30	SAY	Dots in a row. A step is the jump from one dot to the next.
3:38	ASK	How many dots are there? How many steps from the first dot to the last dot? [wait about 30 s for answers]
4:16	CHILDREN	Your turn. A row of three dots. How many steps from the first dot to the last? Show with your fingers. [0:22]
4:38	SLIDE	Show slide **Right and up** (s-rem3)

4:50	**SAY**	A grid of squares. The counter is in the yellow square, at the bottom left.
4:58	**ASK**	Move the counter one right, then one up. Which square is it in? Point to it. [wait about 28 s for answers]
5:34	**CHILDREN**	Your turn. From the middle square, move one more right. Which square? Point. [0:20]
5:54	**CHILDREN**	Say each answer with me: three rows, five in each row; five steps; the middle square. Now the your-turn answers: eight dots, two steps, the middle square in the right-hand column. Then stretch your fingers. [0:22]
	IF STUCK	If a child is unsure how to count steps, touch each dot, then each jump between the dots, with a finger. For right and up, point along the row first, then up the column. For rows and columns, run a finger along a row.
	NOTE	Answers: 3 rows, 5 in each row; 5 steps; the middle square. Your turns: 8; 2 steps; the middle square of the right-hand column. Children answer with a whisper, fingers or hands. Nobody writes. Joining cold: all you need is counting to 10 and the pictures.

8:00 – 12:00 Sneak-peek puzzles

script 3:12 of 4:00

8:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
8:12	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:22	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:34	**READ**	Puzzle one. The counter is on S. It must reach E. Write how far it moves: blank right, blank up.
8:45	**CHILDREN**	Children write on their own paper. [0:40]
9:25	**READ**	Puzzle two. The counter goes from S to E. It moves only right or up. How many different routes can it take? I know because blank.
9:39	**CHILDREN**	Children write or draw their answer. [1:25]
11:04	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: 3 right, 1 up. Puzzle 2 answer: 2 routes (right then up, up then right). Look for good ideas to praise later (a child who drew both routes, or who said right then up and up then right). Do not write scores or keep a list.

12:00 – 23:00 Explore

script 8:36 of 11:00

12:00	**SLIDE**	Show slide **Explore: copy it exactly** (s-explore1)
12:12	**REC**	*Recording: pause now. Copy the house on your copy grid, then walk your button as the directions say. Press play when you have tried.*
12:24	**SAY**	Today the grid page is our tool. Look at the house on the screen. Copy it on the right-hand side of a copy grid, so that it is exactly the same. A ruler or the edge of a card helps with straight lines.
12:46	**CHILDREN**	Children copy the house on a copy grid. Tutor scans the cameras. [2:40]
15:26	**ASK**	How did you copy it exactly? What did you count? [wait about 45 s for answers]
16:16	**SLIDE**	Show slide **A copy and the figure** (s-explore1-ans)
16:24	**SAY**	Here is the figure and an exact copy. Check yours with your finger: is it exactly the same?
16:32	**SLIDE**	Show slide **Explore: walk your button** (s-explore2)
16:44	**SAY**	Round two. Put your button on the bottom left dot of a route grid. Walk your button as I say, and draw the line as you go.
16:58	**READ**	Go two right. Go two up. Go one right. Go one up.
17:05	**CHILDREN**	Children walk the button and draw the line. Wait after each direction. [1:30]
18:35	**ASK**	Where is your button now? [wait about 35 s for answers]

19:12	**SLIDE**	Show slide **The line your button walked** (s-explore2-ans)
19:20	**SAY**	Right, up, right, up. The button is on the top right dot. Check your line.
19:28	**SLIDE**	Show slide **What did we notice?** (s-explore3)
19:36	**ASK**	What did you count when you copied the house? What did you count when you walked your button? [wait about 45 s for answers]
20:30	**SAY**	We count the steps between the dots. Next we say it with numbers.
	IF STUCK	A child who cannot start the copy: find the bottom left corner of the house, then count the steps across and the steps up. A child who puts the button on the wrong dot: it starts on the bottom left dot. A child with no button puts a finger on the dot.
	NOTE	Main room only; you cannot see small drawings, so ask children to say what they counted and to show fingers for the steps. Answers: the house is 2 steps across and 3 steps up; the button ends on the top right dot (3 right, 3 up in all). Do not say the rule before the children do.

23:00 – 34:00 Mini-lesson

script 10:17 of 11:00

23:00	**SLIDE**	Show slide **Steps, not dots** (s-m1)
23:08	**REC**	*Recording: keep watching and copy the examples on your grid page. Rewind if you miss a step.*
23:16	**SAY**	A rectangle on dots. Look at the bottom edge.
23:21	**ASK**	How many dots are on the bottom edge? How many steps long is it? [wait about 40 s for answers]
24:08	**SLIDE**	Show slide **4 dots, 3 steps** (s-m1-ans)
24:16	**SAY**	Four dots, but three steps. We count steps, not dots. A step is the gap between two dots.
24:25	**READ**	Your turn. A row of seven dots. How many steps long is it? Write the number.
24:34	**CHILDREN**	Children write on their own paper. Wait 10 seconds before helping. [0:25]
24:59	**SAY**	Seven dots make six steps. One more dot than steps.
25:04	**SLIDE**	Show slide **Copy corner by corner** (s-m2)
25:12	**SAY**	Copy a figure corner by corner. We start at the bottom left corner, A. We say how far each corner is from the start.
25:24	**ASK**	How many steps right is corner B? How many steps up is corner C? [wait about 40 s for answers]
26:11	**SLIDE**	Show slide **A, B and C** (s-m2-ans)
26:19	**SAY**	B is three right. C is two up. Mark each corner by counting steps from the start. Then join the corners.
26:29	**READ**	Your turn. Copy the triangle on your copy grid. Mark A, then B, then C. Then join them.
26:39	**CHILDREN**	Children copy the triangle. Tutor scans the cameras. [0:45]
27:24	**SLIDE**	Show slide **Say where to go** (s-m3)
27:32	**SAY**	Now we say where to go. We say how many right, then how many up.
27:40	**ASK**	How do we get from S to E? Say right and up. [wait about 35 s for answers]
28:21	**SLIDE**	Show slide **4 right, 2 up** (s-m3-ans)
28:29	**SAY**	Four right, two up. Right and left are opposites. Up and down are opposites.
28:36	**ASK**	Your turn. How do we go from E back to S? [wait about 25 s for answers]
29:06	**SAY**	Four left and two down.
29:09	**SLIDE**	Show slide **Follow the directions** (s-m4)
29:17	**SAY**	Follow the directions on a route grid. Start at S, the bottom left dot. Draw as I say.
29:26	**READ**	Go one right. Go two up. Go three right. Go one up.
29:32	**CHILDREN**	Children draw the line on a route grid. Wait after each direction. [0:45]

30:17	**ASK**	Where do you stop? How many right in all? How many up in all? [wait about 35 s for answers]
30:59	**SLIDE**	Show slide **Where did we stop?** (s-m4-ans)
31:07	**SAY**	We stop at the top right dot. One and three make four right. Two and one make three up.
31:17	**SLIDE**	Show slide **Different routes** (s-m5)
31:25	**SAY**	A route goes only right or up. This route is right, up, up. Let us find another route.
31:34	**ASK**	Is there another route? How can we be sure we have found them all? [wait about 40 s for answers]
32:21	**CHILDREN**	Children draw the routes on route grids. They may rub out and use a grid again. [0:35]
32:56	**SLIDE**	Show slide **Three routes, no more** (s-m5-ans)
33:04	**SAY**	Where does the one right go? First, second or third. That makes three routes and no more. We list them in order so none is missed.
	IF STUCK	If a child counts dots instead of steps, put a finger on a dot and jump to the next dot: one step. If a child mixes left and right, hold up the right hand: right is this way on the screen. If a child cannot find a second route, ask: can the right come first? Can it come last? If a child finds 4 routes, check two are the same.
	NOTE	Key words to use every time: step, right, up, route. Fixed question forms: how many steps? ___ right, ___ up. How many different routes? The slides keep the answers off the screen while you ask: read every question with its numbers aloud. Class numbers are not used in the workbook.

34:00 – 38:00 Brain break

script 3:16 of 4:00

34:00	**SLIDE**	Show slide **Counter break** (s-break)
34:08	**REC**	*Recording: stand up and join in! Follow the directions with your body: small steps right, reach up.*
34:16	**SAY**	Stand up and get a sip of water. I give directions for your body. Small steps right. Reach up high.
34:26	**CHILDREN**	Call directions for the body: two small steps right, one big reach up, one small step left, crouch down. Then call right, right, up, up, and ask where they are on an imaginary grid. Seated children move one arm. [2:50]

38:00 – 70:00 Workbook

script 24:36 of 32:00

38:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
38:08	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
38:17	**READ**	Page 1, Start here. Question 1: Mei draws an L shape on a grid. Part a: copy the L shape on the empty grid. Part b: the L shape is blank steps across and blank steps up. Question 2: a counter starts at S. It moves only right and up to reach E. Part a: the counter moves blank right and blank up. Part b: draw the counter's path from S to E. Question 3: start at S. Go four steps right. Then go one step up. Part a: draw the path. Part b: draw a ring round the dot where you stop.
39:13	**CHILDREN**	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [9:00]
48:13	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
48:21	**READ**	Question 1: a copy of the L shape. Three steps across and three steps up. Question 2: two right, three up, and any path that only goes right or up. Question 3: a line four right then one up, with a ring at the end. Tick your own page with a tick.
48:50	**SAY**	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
48:59	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
49:07	**READ**	Page 2. Question 4: which picture is an exact copy of the T shape? Circle first, second or third. Question 5: three corners of a rectangle are ringed. Part a: draw a ring round the missing corner and join the dots. Part b: the missing corner is blank right and blank up from S. Question 6: tick True or False for each sentence about the path.

49:43	**CHILDREN**	Page 2, questions 4 to 6. [8:20]
58:03	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
58:11	**READ**	Question 4: the second picture. Question 5: three right, two up. Question 6: True, False, True, False. Tick your own page.
58:22	**SLIDE**	Show slide **Choose your next page** (s-choice)
58:30	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
58:46	**MOVE**	Stand and reach up for 30 seconds, then sit down and choose.
59:16	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [3:20]
	IF STUCK	Q1: count the steps across the bottom of the L, then the steps up the side. Q2: count the steps right along the bottom, then the steps up. Q3: put your pencil on S and count four jumps right. Q4: count the steps across the top bar and down the stem in each picture. Q5: the missing corner is above or below one corner and level with another. Q6: check each sentence on the picture.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder has counting the routes on two small grids, a path that repeats and a path with a left move. Go further has adding at the corners (beyond Primary 1 on purpose), a corner the route must pass and drawing a figure for a friend to copy. Do not announce who is on which page. Start here takes a middle child about 17 minutes, so the choice time may be short. Slow class: take the extra minutes from the flex time and do not skip the answer slides.

70:00 – 76:00 Talk time

script 4:26 of 6:00

70:00	**SLIDE**	Show slide **Talk and explain** (s-talk)
70:08	**REC**	*Recording: pause. Tell a grown-up whether two different routes can stop at the same dot, and why.*
70:16	**READ**	Ali goes three right, then two up. Mei goes two up, then three right. Do they stop at the same dot? Do they walk the same number of steps? How do you know?
70:34	**CHILDREN**	Think about it. You may use your grid page and your button. [0:50]
71:24	**CHILDREN**	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]
72:04	**ASK**	Who would like to share? Volunteers first. A pass is always fine. [wait about 100 s for answers]
73:50	**SLIDE**	Show slide **Back to the sneak-peek** (s-sneakans)
73:58	**SAY**	Remember the sneak-peek? Puzzle one: three right, one up. Puzzle two: two routes. Right then up, or up then right.
74:08	**ASK**	Did anyone change their mind while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: yes, they stop at the same dot, and they both walk 5 steps. Both go 3 right and 2 up. Only the order is different. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed. The names in the story are not children in the group.

76:00 – 82:00 Exit ticket

script 4:47 of 6:00

76:00	**SLIDE**	Show slide **Exit ticket** (s-exit)
76:08	**REC**	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:16	**SAY**	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.

76:24	**READ**	E1. A counter starts at S and moves only right and up to reach E. Part a: the counter moves blank right and blank up. E2. A counter moves two right and three up. Then it moves three right and one up. In all it moves blank right and blank up. E3. A figure is five steps across and two steps up. A child draws a copy that is six steps across and two steps up. Part a: is it an exact copy? Circle Yes or No. Part b: an exact copy is blank steps across.
77:16	**CHILDREN**	Children answer E1 to E3 on their own. [3:10]
80:26	**SLIDE**	Show slide **Check the exit ticket** (s-exitans)
80:34	**READ**	E1: two right, two up. E2: five right, four up. E3: No. Five steps across.
80:43	**SAY**	Tick your own ticket with a tick. Well done.
	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result.

82:00 – 85:00 Look-back

script 2:06 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:08	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:14	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
82:46	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:36 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:08	**SAY**	Well done, inquirers. Next lesson is Cubes and Views. Please have 12 small blocks of one size, like building bricks or dice. If you have no blocks, tell us: the lesson works with pictures. Also have the 4 paper squares from your book, cut out by a grown-up. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: add at the corners** (s-opt) *(optional)*
	SAY	Optional, only if you are on time. The boxes show how many routes reach each corner. The bottom row and the left column are all one. What goes in the empty box? It is the box on its left plus the box below it.
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: the empty box is 1 + 2 = 3. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Does the order of the moves matter?	4 (optional)	A child says: if we do the same moves in a different order, we stop at the same dot. Is that always true? Test it with 2 right and 1 up, in two orders. Draw both lines on a route grid.	*Recording: pause. Test whether the same moves in a different order stop at the same dot.*	Yes. The total moves right and up are the same, so we stop at the same dot. The two lines are different.
Make a route puzzle for your grown-up	4 (optional)	Draw a small grid on paper. Ask your grown-up how many routes there are from the bottom left to the top right, moving only right or up. Then check by listing the routes in order.	*Recording: pause. Draw a small grid and ask a grown-up to count the routes. Check by listing.*	No single answer. Check by listing every route with R and U in order, so none is missed.
Teach it back	3 (optional)	Pretend a toy counts the dots in a row and says that is the number of steps. Show the toy how to count the jumps between the dots instead.	*Recording: pause. Show a grown-up how to count the steps in a row of dots.*	A row of 5 dots has 4 steps. A step is a jump from one dot to the next, so there is one more dot than steps.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): The counter is on S. It must reach E. Write how far it moves: ___ right, ___ up.

Answer: 3 right, 1 up.

Puzzle 2 (think): The counter goes from S to E. It moves only right or up. How many different routes can it take? I know because ___.

Answer: 2 routes: right then up, or up then right.

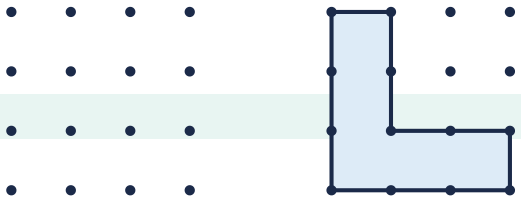
Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Mei draws an L shape on a grid.**

(a) Copy the L shape on the empty grid.

(b) The L shape is ☐ steps across and ☐ steps up.



Answer: A copy of the L shape; 3; 3

Hint (say this first): Count the jumps between the dots along the bottom. Then count up the side.

Worked solution: The L shape is 3 steps across and 3 steps up. To copy it, start at the top left dot of the empty grid and draw: 1 right, 2 down, 2 right, 1 down, 3 left and 3 up. The copy has the same six corners in the same places on its own grid.

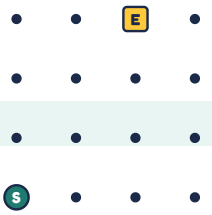
Common mistakes: Counting dots, not steps (4 and 4).; Making the bottom bar too short or too long by one step.

Time for a middle child: about 3.5 min.

S2 **A counter starts at S. It moves only right and up to reach E.**

(a) The counter moves ☐ right and ☐ up.

(b) Draw the counter's path from S to E.



Answer: 2 right, 3 up; any path that goes only right and up

Hint (say this first): Count the steps along the bottom, then the steps up.

Worked solution: E is 2 steps to the right of S and 3 steps up. One path is 2 right and then 3 up. Any path with 2 right steps and 3 up steps works, in any order.

Common mistakes: Counting dots, not steps (3 right).; Swapping right and up (3 right, 2 up).

Time for a middle child: about 2.5 min.

S3 **Start at S. Go 4 steps right. Then go 1 step up.**

(a) Draw the path.

(b) Draw a ring round the dot where you stop.



Answer: A line 4 right and then 1 up; ring on the dot where it stops

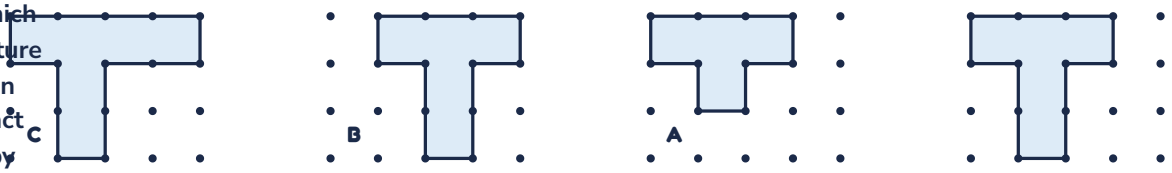
Hint (say this first): Put your pencil on S. Jump to the next dot 4 times.

Worked solution: Start on S. 4 steps right: count 1, 2, 3, 4 jumps along the bottom row. Then 1 step up. The ring goes on the dot 4 right and 1 up from S.

Common mistakes: Counting S as the first step.; Going 4 up and 1 right (swapped).

Time for a middle child: about 3 min.

S4 **Which picture is an exact copy of the T shape?**

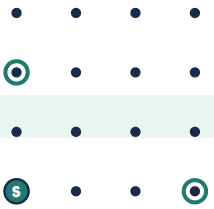


Circle A, B or C.

Answer: B

Hint (say this first): Count the steps across the top bar and down the stem in the T shape. Then count in each picture.
Worked solution: The T shape has a top bar 3 steps across and a stem that makes it 3 steps tall. Picture A has a stem 1 step too short. Picture C has a top bar 1 step too wide. Picture B is the same T shape moved 1 step to the right, so B is the exact copy.
Common mistakes: Choosing the picture that sits in the same place as the T (a copy can be in a different place).; Counting dots instead of steps.
Time for a middle child: about 2.5 min.

S5 **Three corners of a rectangle are ringed. Join the dots to finish the rectangle.**



(a) Draw a ring round the missing corner.
(b) The missing corner is ☐ right and ☐ up from S.

Answer: 3 right, 2 up

Hint (say this first): The missing corner is level with one ringed corner and above another.
Worked solution: One ringed corner is 3 steps right of S. The other is 2 steps up from S. The missing corner has both moves: 3 right and 2 up. Join the four corners to make the rectangle.
Common mistakes: Putting the corner in the same row or column as S.; Swapping right and up (2 right, 3 up).
Time for a middle child: about 3 min.

True/False statements: The path goes 5 steps right.; The path goes 2 steps up.; The bottom line touches 5 dots.; E is 5 right and 1 up from S.

S6 **Look at the path from S to E. Tick True or False for each sentence.**



Answer: True; False; False; True

Hint (say this first): Count the jumps along the path. Do not count the dots.
Worked solution: The path goes 5 steps right and 1 step up. The bottom line touches 6 dots, because a line of 5 steps has 6 dots. So: 5 steps right is true; 2 steps up is false (1); 5 dots is false (6); E is 5 right and 1 up is true.
Common mistakes: Ticking True for the number of dots.; Counting the starting dot as a step.
Time for a middle child: about 3 min.

Think harder (page 3): the taught idea in a new setting

T1 **A counter goes from S to E. It moves only right or up. The grid is 3 steps across and 1 step up.**



(a) One route is R R R U. R means right and U means up. Write the other routes: ____ ____ ____
(b) There are ☐ routes in all.

Answer: U R R R, R U R R and R R U R (any order); 4 routes

Hint (say this first): Every route has 3 R and 1 U. Where can the U go?
Worked solution: Every route has 3 right steps and 1 up step. The U can be first, second, third or last: U R R R, R U R R, R R U R and R R R U. That is 4 routes.
Common mistakes: Writing the same route twice.; Stopping at 3 routes and missing one.
Time for a middle child: about 5 min.

T2 **A counter goes from S to E. It moves only right or up. The grid is 2 steps across and 2 steps up.**



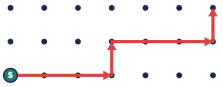
(a) Three routes start with U: U U R R, U R U R and U R R U. Write the routes that start with R: ____ ____ ____

(b) There are ☐ routes in all.

Answer: R R U U, R U R U and R U U R (any order); 6 routes

Hint (say this first): Each route has 2 R and 2 U. Start with R, then find where the other R goes.
Worked solution: Routes that start with R: R R U U, R U R U and R U U R. Routes that start with U are U U R R, U R U R and U R R U. 3 and 3 make 6 routes. Each route has 2 R and 2 U; the two R can sit in 6 pairs of places.
Common mistakes: Writing a route that starts with U again.; Counting 4 routes.
Time for a middle child: about 5 min.

T3 **Ravi's counter walks 3 right and 1 up, again and again. The picture shows 2 sets.**



(a) After 3 sets, the counter is ☐ right and ☐ up from S.

(b) After 5 sets, the counter is ☐☐ right and ☐ up from S.

Answer: 9 right, 3 up; 15 right, 5 up

Hint (say this first): Each set adds 3 right and 1 up. Count in threes.
Worked solution: One set is 3 right and 1 up. 2 sets: 6 right and 2 up. 3 sets: 9 right and 3 up. 5 sets: 3, 6, 9, 12, 15 right and 5 up.
Common mistakes: Adding 3 right only once.; Forgetting the steps up.
Time for a middle child: about 5 min.

T4 **Start at S. Go 3 steps right. Go 2 steps up. Then go 1 step left.**



(a) Draw the path.

(b) You stop ☐ right and ☐ up from S.

Answer: 2 right, 2 up

Hint (say this first): Left is the opposite of right. One step left undoes one step right.
Worked solution: 3 right then 1 left leaves 2 right of S. The 2 up stay. You stop 2 right and 2 up from S.
Common mistakes: Writing 3 right (forgetting the left step).; Moving 1 step right instead of left.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **The boxes show how many routes reach each corner. S is the bottom left box. E is the top right box.**



(a) Fill in the empty boxes.

(b) There are ☐☐ routes from S to E.

Answer: 4 (middle row); 3, 6 and 10 (top row); 10 routes

Hint (say this first): Look at the filled boxes. How is each one made from the box on its left and the box below it?
Worked solution: Each box is the box on its left plus the box below it. Middle row: 3 + 1 = 4. Top row: 1 + 2 = 3, then 3 + 3 = 6, then 6 + 4 = 10. So 10 routes reach E on a grid 3 across and 2 up.
Common mistakes: Writing 1 in every empty box.; Adding the box above instead of the box below.
Time for a middle child: about 6 min.

G2 **A counter goes from S to E, moving only right or up. The grid is 3 steps across and 2 steps up. The route must pass the ringed corner.**

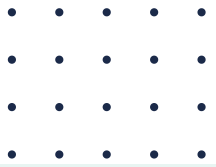


(a) Routes from S to the ringed corner: ☐
(b) Routes from the ringed corner to E: ☐
(c) Routes from S to E that pass the ringed corner: ☐

Answer: 3; 2; 6

Hint (say this first): Each way to the ringed corner can be followed by each way on to E.
Worked solution: To the ringed corner: 2 right and 1 up gives 3 routes (R R U, R U R, U R R). From it to E: 1 right and 1 up gives 2 routes (R U, U R). Each of the 3 first parts can be followed by either of the 2 second parts: 3 groups of 2 is 6 routes.
Common mistakes: Adding 3 and 2 to get 5.; Counting all 10 routes on the grid.
Time for a middle child: about 6 min.

G3 **Draw a shape with 4 corners on the grid. A is its bottom left corner. Many answers are right.**



(a) B is ☐ right and ☐ up from A.
(b) C is ☐ right and ☐ up from A.
(c) D is ☐ right and ☐ up from A.

Answer: Many answers are right: three more corners, each said as moves right and up from A, that join A to make a shape

Hint (say this first): Mark A first. Then pick a dot, count the steps right and the steps up, and write them.
Worked solution: For example, a rectangle: B is 3 right and 0 up, C is 3 right and 2 up, D is 0 right and 2 up. A friend can copy the shape by marking each corner from A and joining the corners.
Common mistakes: Counting dots, not steps.; Giving a corner that is left of A.
Time for a middle child: about 5 min.

Investigation (page 5): Same steps, different routes

Easy first step: Draw two different routes from S to E, one on each grid. Count the steps in each route: ____ steps and ____ steps.
Then: Do all the routes from S to E have the same number of steps? Write why.

Model answer: Both routes have 5 steps. Every route from S to E needs 3 steps right and 2 steps up, so every route has 5 steps. Only the order of the steps changes. (There are 10 routes on this grid.)

Look for: A child who says every route needs the same moves: 3 right and 2 up, only in a different order. Credit any clear reason. A child who draws two routes and counts 5 and 5 has the first step.
Common mistakes: Counting dots (6) instead of steps.; Saying a route with more turns is longer.

Talk and explain (minutes 70 to 76)

Ali goes 3 right, then 2 up. Mei goes 2 up, then 3 right. Do they stop at the same dot? Do they walk the same number of steps? How do you know? Sentence starter: I think ... because ...

Model answer: Yes, they stop at the same dot, and they both walk 5 steps. Both go 3 right and 2 up. Only the order is different, so the lines are different but the start and the end are the same.

Look for: A child who says the moves are the same, only in a different order. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **A counter starts at S and moves only right and up to reach E.**



(a) The counter moves ☐ right and ☐ up.

Answer: 2 right, 2 up

Worked solution: E is 2 steps right of S and 2 steps up. Count the jumps, not the dots.
Common mistakes: Counting dots (3 and 3).; Swapping right and up.

E2 **A counter moves 2 right and 3 up. Then it moves 3 right and 1 up.**

(a) In all it moves ☐ right and ☐ up.

Answer: 5 right, 4 up

Worked solution: 2 and 3 make 5 right. 3 and 1 make 4 up.
Common mistakes: Adding the rights and the ups together (9).; Writing 3 right and 1 up (the last move only).

E3 **A figure is 5 steps across and 2 steps up. A child draws a copy that is 6 steps across and 2 steps up.**

(a) Is it an exact copy? Circle Yes or No.
(b) An exact copy is ☐ steps across.

Answer: No; 5

Worked solution: An exact copy is the same size. 6 steps across is 1 step too wide, so it is not an exact copy. The copy should be 5 steps across.
Common mistakes: Saying Yes because the height is the same.; Writing 6 for part b.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (counting steps, not dots; copying exactly; counting the routes; something else); *What helped me* (the grid page; my button; drawing the routes; I was not stuck); *Next time I will try* (counting steps; saying where to go; finding every route; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Ben's counter moves 4 right and 1 up. Then it moves 2 right and 3 up. In all it moves ___ right and ___ up.	6 right, 4 up
H2	Start here	A row has 8 dots. How many steps is it from the first dot to the last dot? It is ___ steps.	7
H3	Start here	A line goes from the 1st dot to the 5th dot in a row. Is it 5 steps long? Circle Yes or No. It is ___ steps long.	No. 4 steps.
H4	Think harder	A grid is 1 step across and 4 steps up. A counter goes from the bottom left to the top right. It moves only right or up. How many different routes are there? There are ___ routes.	5
H5	Start here	Mei has 7 red marbles and 7 blue marbles. How many marbles does she have altogether? She has ___ marbles.	14

Online practice (`out/l33/practice`, key `t4l33`): 13 items; 9 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to say where things are on a grid and to copy a figure exactly. This week: copying on a dot grid, saying right and up, and counting routes.

Term line (plan, section 5): Your child is learning to choose a way to solve a problem, and to make and test their own puzzles. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Copy on the Grid** s-title	TODAY WE WILL: / Copy a figure exactly. / Say where to go on a grid. / Count the different routes.	Lesson 3 of Term 4, 90 minutes. Objective: copy a figure on a dot grid by counting steps (not dots), say a move as 'right and up', follow directions, and list every route that only goes right or up. Plain statement for families: copying a figure on a dot or square grid and following 'right' and 'up' directions are Primary 1 work; counting routes (how many different ways), adding at the corners and a route that must pass a corner go past the school Primary 1 syllabus on purpose. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 23, mini-lesson 23 to 34, brain break 34 to 38, workbook 38 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Say the recording notice before you start the recording, in one line: you, the shared screen and any unmuted voice; a child may keep the camera off.
2	2–8	**Warm-up: rows and columns** s-rem1	[picture: array] / Rows go across. Columns go down. How many rows? How many in each row?	Rows and columns (Term 2 Lesson 5, taught again). Rows go across, like lines of writing. Columns go down, like the legs of a table. Ask: how many rows (3), how many counters in each row (5). A child who is unsure points along each row with a finger. Your turn, no picture: 2 rows with 4 dots in each row; how many dots in all (8). Children answer aloud. No writing.
3	2–8	**Steps between dots** s-rem2	[picture: grid] / A step is the jump from one dot to the next. How many steps from the first dot to the last dot?	Steps between dots (Term 1 Lesson 6, jumps, taught again). The picture shows a row of 6 dots. A step is the jump from one dot to the next, so from the first dot to the last dot there are 5 steps: one fewer than the number of dots. Ask: how many dots (6), how many steps (5). A child who is unsure touches each dot, then each jump, with a finger. Your turn, no picture: a row of 3 dots; how many steps from the first dot to the last (2). Children answer aloud. No writing.
4	2–8	**Right and up** s-rem3	[picture: grid] / The counter is in the yellow square. Move it 1 right, then 1 up. Which square?	Right and up on a grid of squares (Term 1 Lesson 11, position words, taught again). The counter is in the bottom left square (yellow). Right is along the row, up is up the column. Move 1 right and then 1 up: the counter is in the middle square. Ask: which square is it in (the middle square). A child who is unsure moves a finger along the picture. Your turn: from the middle square move 1 more right; which square (the middle square of the right-hand column). Children answer aloud by pointing or saying. No writing.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / The counter is on S. / It must reach E. / Write: ___ right, ___ up. / [picture: grid] / PUZZLE 2: think: / The counter goes from S to E. / It moves only right or up. / How many different routes? I know because ___. / [picture: grid]	Puzzle 1 is an access question: S is at the bottom left, E is 3 dots to the right and 1 up, so the answer is 3 right, 1 up. Puzzle 2 is a thinking question on the lesson idea: on a grid 1 step across and 1 step up the counter can go right then up, or up then right: 2 routes. A child who writes 'right then up' and 'up then right' has done well. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: 3 right, 1 up; 2 routes. Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.

6	12–23	**Explore: copy it exactly** s-explore1	[picture: grid] / Copy the house on your copy grid. Make it exactly the same.	Round 1, about 5 minutes. Each child has the grid page (page 6 of the book) and a pencil. The slide shows a house on a dot grid, 2 steps across and 3 steps up. Children copy it on the right-hand side of a copy grid (the grid with the dashed line); they may use a ruler or the edge of a card for straight lines. Walk the cameras. The picture is not shown with its copy until the next slide. Ask: how did you copy it exactly; what did you count. Listen for counting the dots (a copy that is a step too wide) and for counting the steps. Do not give the rule yet. A child with no printed grid draws rows of dots on paper: 4 dots across, 4 dots down. Nobody holds a drawing up to the camera: children check their own copy on the next slide.
7	12–23	**A copy and the figure** s-explore1-ans	[picture: grid] / The figure / [picture: grid] / An exact copy	Children check their own copy against the picture. Both pictures are 2 steps across and 3 steps up, with the point of the roof on the middle dot of the top row. Say: put your finger on the bottom left corner of the house and count the steps along the bottom, then up the side. Children who were one step too wide counted the dots, not the steps: say that is what we will look at in the mini-lesson. Do not name any child. Ticks only, no marks.
8	12–23	**Explore: walk your button** s-explore2	Put your button on the bottom left dot of a route grid. / Go 2 right. / Go 2 up. / Go 1 right. / Go 1 up. / Walk the button and draw the line as you go.	Round 2, about 4 minutes. Each child has a route grid on the grid page and a button (or small coin) as the counter. Read the four directions one at a time, slowly, and wait for children to move and draw before the next: go 2 right, go 2 up, go 1 right, go 1 up. Then ask where the button is now. The button ends on the top right dot (3 right and 3 up in all). Walk the cameras: you can see the counters on the table. The line is drawn on the next slide. Nobody holds a drawing up to the camera. A child without a button puts a finger on the dot, or makes a small cross with a pencil.
9	12–23	**The line your button walked** s-explore2-ans	[picture: grid] / Right, up, right, up. The button is on the top right dot.	Children check their own line. The path goes 2 right, then 2 up, then 1 right, then 1 up and ends on the top right dot E. In all that is 3 right and 3 up. Say: we said the moves in order, and the order matters for the line, but look at where the button stopped. Do not ask about other routes yet. Ticks only.
10	12–23	**What did we notice?** s-explore3	What did you count when you copied the house? / What did you count when you walked the button?	Take one or two answers from volunteers. Key ideas: we counted steps between the dots (not the dots); the dots are the places and the steps are the jumps; the directions are right and up. If a child says we counted dots, ask: how many dots are in the bottom row of the house, and how many steps long is it (3 dots, 2 steps). Repeat each child answer in your own words so the recording picks it up. Do not give the rule before the children say it; the mini-lesson makes it exact.
11	23–34	**Steps, not dots** s-m1	[picture: grid] / How many dots are on the bottom edge? How many steps long is it?	Worked example. The rectangle's bottom edge touches 4 dots, but it is 3 steps long. Ask: how many dots on the bottom edge (4), how many steps long is the edge (3). The answer is on the next slide. Say: a step is the gap between two dots, so we count the gaps. The edge has one more dot than steps. Then the your-turn question: a row of 7 dots, how many steps long is it (6). Children write on their own paper.
12	23–34	**4 dots, 3 steps** s-m1-ans	[picture: grid] / The bottom edge touches 4 dots. It is 3 steps long. We count steps, not dots.	The answer for the example on the slide before. The arrows are the steps along the bottom edge: 3 steps, 4 dots. Read it aloud so that children can tick their own work, and say what the picture showed. Your-turn answer: a row of 7 dots is 6 steps long.
13	23–34	**Copy corner by corner** s-m2	[picture: grid] / Corner A is the start, at the bottom left. How far right is corner B? How far up is corner C?	Worked example. A triangle with three corners. Start at the bottom left corner A. Corner B is on the same row, 3 steps to the right. Corner C is straight above A, 2 steps up. Ask: how many steps right is B (3); how many steps up is C (2). The answer is on the next slide. Say: to copy a figure, mark each corner by counting steps from the start, then join the corners. Then the your-turn: copy the triangle on your copy grid: mark A, mark B, mark C, join.

14	23–34	**A, B and C** s-m2-ans	A: the start. / B: 3 right. / C: 2 up. / [picture: grid]	The answer for the example on the slide before. A is the start. B is 3 steps right of A. C is 2 steps up from A. Then join A to B, B to C and C to A. Read it aloud so that children can tick their own copy, and say what we counted. The line from C to B slants: it goes from dot to dot like the others.
15	23–34	**Say where to go** s-m3	[picture: grid] / How do we get from S to E? ___ right, ___ up.	Worked example. S is at the bottom left and E is at the top right of a grid 4 steps across and 2 steps up. We say a move as how many right and how many up: 4 right, 2 up. Ask: how do we get from S to E (4 right, 2 up). The answer is on the next slide. Say: right, then up. Accept 2 up then 4 right as well. Then the your-turn: how do we get from E back to S: 4 left and 2 down.
16	23–34	**4 right, 2 up** s-m3-ans	4 right, 2 up. To come back: 4 left, 2 down. / [picture: grid]	The answer for the example on the slide before. 4 right, 2 up. Left and right are opposites and up and down are opposites, so the way back is 4 left and 2 down. Read it aloud so that children can tick their own work. A child who says 6 steps has counted every step: that is the total length of the way, and it is also right; the move is said in two parts, right and up.
17	23–34	**Follow the directions** s-m4	Start at S. / Go 1 right. / Go 2 up. / Go 3 right. / Go 1 up. / [picture: grid]	Worked example. Children use a route grid and draw as you read: go 1 right, go 2 up, go 3 right, go 1 up. Read one direction at a time and wait. Ask: where do you stop; how many right in all; how many up in all. The line and the answer are on the next slide: the path ends at the top right dot, 4 right and 3 up in all. Children tick their own work.
18	23–34	**Where did we stop?** s-m4-ans	[picture: grid] / We stop at the top right dot. In all: 4 right and 3 up.	The answer for the example on the slide before. 1 + 3 = 4 right in all, 2 + 1 = 3 up in all. The path ends at E. Read it aloud so that children can tick their own work. Say: we can add the rights and add the ups to see where we stop.
19	23–34	**Different routes** s-m5	[picture: grid] / A route goes only right or up. This route is R, U, U. Find another route.	Worked example. The counter goes from S to E and may only move right (R) or up (U). The grid is 1 step across and 2 steps up, so every route has one R and two U. The picture shows R U U. Ask: is there another route; how can we be sure we have found them all. Give time, then show the three routes on the next slide. The way to be sure: the single R can be first, second or third. Children draw the routes on route grids; they may rub out and use the same grid again. A child may write the letters instead of drawing.
20	23–34	**Three routes, no more** s-m5-ans	[picture: grid] / R U U / [picture: grid] / U R U / [picture: grid] / U U R / The R can go first, second or third. That makes 3 routes.	The answer for the example on the slide before. R U U, U R U and U U R. Say: where does the one R go: first, second or third. That gives 3 routes and no more, and listing them in order means none is missed. Read it aloud so that children can tick their own work. Do not give the pattern for bigger grids: that is Think harder. If a class is quick, one volunteer may say how many routes there would be if the grid were 1 across and 3 up (4: the R can be in 4 places).
21	34–38	**Counter break** s-break	Stand up. Follow my directions with your body. / Take small steps right. Reach up high. / Sip some water.	Four minutes. Stand up on camera with a little clear space. Call directions for the body: two small steps to the right, one big reach up, one small step to the left, crouch down, and so on. Then call a short route with the body: right, right, up, up and ask children to say where they are on an imaginary grid. Say right and up clearly, and use the screen as the front so left and right match what children see on camera. No equipment, no running. A child may stay seated and move one arm. Do not give prizes or compare children.
22	38–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Take your time. / questions S1, S2, S3	Read questions 1 to 3 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. Question 1: copy the L shape on the empty grid, then say how many steps across and up. Question 2: from S to E, how many right and how many up, and draw a path. Question 3: follow the directions and ring where you stop.

23	38–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1: a copy of the L, 3 and 3. Q2: 2 right, 3 up, and any path that only goes right or up. Q3: a line 4 right then 1 up, ringed at the end.
24	38–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. Take your time. / questions S4, S5, S6	Read questions 4 to 6 aloud. Question 4: which picture is an exact copy of the T; circle first, second or third. Question 5: three corners of a rectangle are ringed; draw the missing corner and say how far it is from S. Question 6: tick True or False for four sentences about the path. Wait 10 seconds before helping.
25	38–70	**Check page 2** s-ws2ans	answers S4, S5, S6	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q4 second. Q5 the corner is 3 right and 2 up from S. Q6 True, False, True, False.
26	38–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Count the routes, and a path that repeats / Go further (page 4) | Add at the corners, a corner the route must pass, your own figure / Investigation (page 5) | Same steps, different routes / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
27	70–76	**Talk and explain** s-talk	TWO CHILDREN WALK FROM S: / Ali goes 3 right, then 2 up. / Mei goes 2 up, then 3 right. / Do they stop at the same dot? Do they walk the same number of steps? / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Volunteers first; a pass is always fine; never call a child by name to answer. Model answer: yes, they stop at the same dot, and both walk 5 steps. Both go 3 right and 2 up; only the order is different. The routes are different but the start and the end are the same. Link to Explore and the Investigation: different routes, the same number of steps. Repeat each child answer in your own words for the recording. The names in the story are for the story only: nobody in the group is a mover.
28	70–76	**Back to the sneak-peek** s-sneakans	Puzzle 1: 3 right, 1 up. Puzzle 2: 2 routes. Right then up, or up then right.	Reveal both puzzles now. Say: many of you wrote a good answer. Do not read out anyone earlier answer. Puzzle 1: S to E is 3 right and 1 up. Puzzle 2: on a grid 1 across and 1 up there are 2 routes: right then up, and up then right. It was the same idea as the mini-lesson: say where to go, and list every route so none is missed.
29	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud, with the numbers (the figure on the slide is small). Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
30	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: E1 2 right, 2 up; E2 5 right, 4 up; E3 No, then 5 steps across.
31	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.

32	85–90	**Well done, inquirers!** s-end	Show a grown-up how to walk from S to E on your grid page. / NEXT LESSON: / Cubes and Views / Have ready: 12 small blocks of one size (bricks or dice). The 4 paper squares from your book, cut out by a grown-up.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that the next lesson needs 12 small blocks of one size (building bricks or dice; no blocks at home is fine, the lesson works with pictures) and the 4 paper squares from the fold-squares page of Book 7, torn out along the dashed line and cut out by a grown-up beforehand. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide.
33	optional	**Go further: add at the corners** s-opt	[picture: grid] / Each box says how many routes reach that corner. The empty box is the box on its left plus the box below it.	Optional, only if you are on time. The boxes stand for the corners of a grid 1 step across and 2 steps up, with S in the bottom left box. Each box says how many routes reach that corner. The bottom row and the left column are 1 (only one way to get there). The empty top right box is the box on its left plus the box below it: 1 + 2 = 3 routes, the same 3 routes as in the mini-lesson. Ask: what goes in the empty box (3). Take two answers and move on. This links to counting the routes: adding the two corners before it gives the number of routes to a corner.
34	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Does the order of the moves change where we stop? / Make a puzzle | Draw a grid puzzle for your grown-up. / Teach it back | Show a toy how to count steps, not dots. / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook, your grid page and a button. Pause or rewind any time.*	Gather your workbook, grid page, button, pencil and paper. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Say your answers out loud before I say them. Pause if you need longer.*	Play along. Say each answer out loud before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Copy the house on your copy grid, then walk your button as the directions say. Press play when you have tried.*	Pause. Copy the house on a copy grid. Walk your button along the directions and draw the line.
23:00	Mini-lesson	*Recording: keep watching and copy the examples on your grid page. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
34:00	Brain break	*Recording: stand up and join in! Follow the directions with your body: small steps right, reach up.*	Stand up. Move as the directions say, then sit down.
38:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up whether two different routes can stop at the same dot, and why.*	Pause. Explain your answer to someone at home and show how you know.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.

82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up how to walk from S to E on your grid page.*	Show a grown-up your grid page and walk the button for them.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:36	1:48	ok
remember	6:00	4:16	5:04	ok
sneak	4:00	3:12	3:45	ok
explore	11:00	8:36	10:12	ok
mini	11:00	10:17	12:07	OVER (slow class)
break	4:00	3:16	3:54	ok
work	32:00	24:36	29:15	ok
talk	6:00	4:26	5:16	ok
exit	6:00	4:47	5:36	ok
look	3:00	2:06	2:27	ok
flex	5:00	0:36	0:40	ok
Total	90:00	67:43	80:03	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 9.0 min for page 1 and 8.5 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.